Photodiagnosis and Photodynamic Therapy 55 (2025) 104774

Contents lists available at ScienceDirect

Photodiagnosis and Photodynamic Therapy

journal homepage: www.elsevier.com/locate/pdpdt

Approach to optic disc swelling with macular subretinal fluid and choroidal folds: a narrative review

ARTICLE INFO

Keywords:
Optic disc oedema
Papilloedema
Scleritis
Central serous chorioretinopathy
Retina
Ophthalmology

Optic disc swelling accompanied by macular subretinal fluid (SRF) and choroidal folds has several differential diagnoses (DDx). When the presentation is unilateral and associated with painful eye injection ("hot eye"), posterior scleritis (PS) [1], thyroid eye disease (TED) [2], and idiopathic orbital inflammatory disease (IOID) [3] represent the most likely DDx, while central serous chorioretinopathy (CSCR) is less probable [1]. Photodiagnosis using optical density ratio (ODR) of SRF to other intraocular structures on optical coherence tomography may help differentiate the aetiologies. Jiangbolati et al. recently reported an interesting case of PS that was misdiagnosed as CSCR but responded successfully to high-dose systemic corticosteroid [1]. The 39-year-old man presented with unilateral disc swelling, macular SRF, and choroidal folds in a "hot eye" [1]. However, a favourable response to high-dose systemic corticosteroid does not rule in PS alone, as a similar response may occur in TED and IOID. Furthermore, the absence of proptosis in TED is not uncommon, as euthyroid Graves' ophthalmopathy (a cold presenter) has been recently demonstrated in Asian populations [4]. Additional diagnostic confirmation would ideally include thyroid function tests and orbital imaging - investigations not documented in this case report - to help rule out TED and IOID.

The diagnostic workup for optic disc swelling necessitates differentiation between true versus pseudo-disc swelling (PSD) [5]. PSD may result from myelinated retinal nerve fibre layer (RNFL) at the disc, high myopia with tilted disc, small discs, optic nerve head drusen [5,6], or genetic disorders such as Leber's hereditary optic neuropathy [7]. B-scan ultrasonography or fundus autofluorescence imaging may help identify the aetiologies [5], such as revealing buried drusen over the optic nerve head. True optic disc swelling can be confirmed by fundus fluorescein angiography, which reveals leakage from the disc [8]. Bilateral disc swelling may result from malignant hypertension, Vogt-Koyanagi-Harada disease (VKH), and elevated intracranial pressure (papilloedema), etc [9].

Distinguishing papilloedema (bilateral true optic disc swelling caused by raised intracranial pressure [ICP]) from pseudo-papilloedema (bilateral optic disc swelling caused by malignant hypertension, VKH etc.) hinges critically on confirming elevated ICP. While clinical features like obscuration of disc margins, haemorrhages, exudates, and cotton wool spots strongly suggest genuine papilloedema, definitive differentiation from pseudo-papilloedema often requires objective ICP measurement. The gold standard for accurate, continuous ICP monitoring is the placement of an intraventricular catheter (IVC). This technique involves drilling a small burr hole through the skull, inserting the catheter into the lateral ventricle, and directly measuring cerebrospinal fluid (CSF) pressure. Alternatively, an external ventricular drain (EVD) serves the same purpose of monitoring CSF pressure. While less invasive methods can provide supportive evidence, they lack the absolute accuracy of direct ventricular measurement. Transcranial Doppler ultrasound (TCD) indirectly assesses ICP by measuring middle cerebral artery blood flow velocity and calculating the pulsatility index, which correlates with ICP. Optic nerve sheath diameter (ONSD) measurement via orbital ultrasound utilises the optic nerve sheath's distension under elevated CSF pressure. While TCD and ONSD serve as valuable non-invasive screening tools, only IVC- or EVD-confirmed elevated ICP definitively establishes genuine papilloedema; normal ICP strongly supports pseudo-papilloedema.

If optic disc swelling is just observed in one eye, the DDx for unilateral disc swelling is considerably more extensive (Fig. 1). The initial diagnostic approach involves distinguishing between "hot eye" and "white eye" presentation. "Hot eye" cases suggest PS, TED, IOID, or carotid-cavernous fistula (CCF), whereas "white eye" cases may be further categorised based on the extent of ocular signs (Fig. 1).

In "hot eye" presentations, differentiating between PS, TED, IOID, and CCF proves clinically challenging due to overlapping features including proptosis, pain, and diplopia. A systematic approach combining clinical evaluation, imaging, laboratory tests, and specialised procedures is essential for accurate diagnosis, with detailed history taking constituting the first and most fundamental step. Severe orbital pain characterises PS and IOID, whereas TED commonly presents with milder discomfort or irritation. Systemic hyperthyroidism symptoms suggest TED, while autoimmune associations (e.g., rheumatoid arthritis, vasculitis) point to PS. Trauma may precipitate both direct and indirect types of CCF; therefore, head trauma screening should be conducted in addition to evaluating intracranial inflammatory or infectious history. IOID remains a diagnosis of exclusion, typically lacking an identifiable systemic cause.

Physical examinations should follow history taking. Eyelid retraction

https://doi.org/10.1016/j.pdpdt.2025.104774
Received 9 July 2025; Received in revised form 6 August 2025; Accepted 20 August 2025
Available online 20 August 2025
1572-1000/© 2025 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

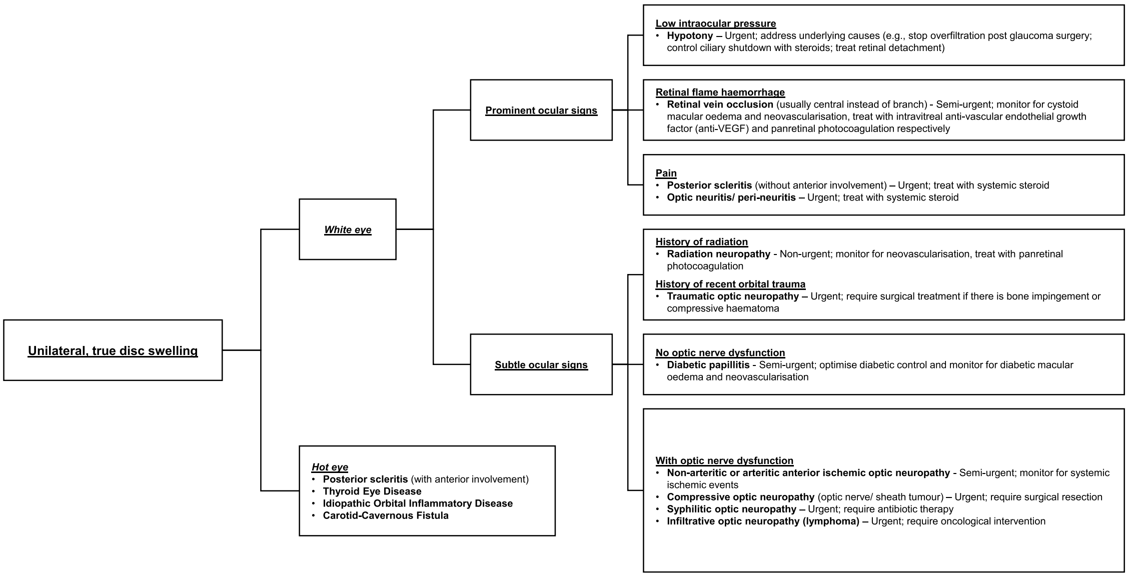

Fig. 1. Approach to unilateral true optic disc swelling.

and conjunctival injection are characteristic of TED, whereas chemosis (conjunctival swelling) often occurs in CCF or IOID. Assessment of extraocular muscles (EOM) movement restriction and diplopia is essential, as these findings are common in TED, IOID, and CCF but less prominent in PS except in severe cases. Pulsatile proptosis or an audible orbital bruit strongly suggests CCF, serving as key distinguishing features from other aetiologies.

Imaging is critical for identifying orbital structural and vascular abnormalities, particularly computed tomography (CT) and magnetic resonance imaging (MRI). In TED, typical findings include bilateral, symmetrical, tendon-sparing EOM enlargement – most commonly affecting the inferior rectus, followed by the medial rectus. A hallmark distinguishing feature of TED versus IOID is tendon-sparing muscle enlargement. While TED commonly presents with proptosis and increased orbital fat volume, IOID often demonstrates inflammation extending to the lacrimal gland (dacryoadenitis).

In CCF, dilated superior ophthalmic veins and abnormal cavernous sinus fullness are pathognomonic findings. Cerebral angiography remains the diagnostic gold standard, confirming both the presence and classification (direct/indirect) of the fistula. In contrast, ultrasonography serves as a highly effective diagnostic tool for PS, demonstrating scleral thickening and Tenon's capsule involvement (T-sign). It can additionally detect associated findings such as subretinal fluid or choroidal detachment. Blood tests support TED diagnosis through thyroid hormone levels or specific antibody testing. Inflammatory markers (e.g., erythrocyte sedimentation rate or C-reactive protein) are often elevated in IOID or PS. For clear cases, IOID may require biopsy to exclude malignancy or lymphoproliferative disorders.

In "white eye" presentations, cases can be sub-classified by prominent versus subtle ocular signs. Prominent ocular signs include: (1) abnormal intraocular pressure on Goldmann applanation tonometry, (2) flame-shaped retinal haemorrhages observed via ophthalmoscopy, or (3) pain at rest or during EOM movement testing. Subtle ocular signs may include: (1) visual field defects detectable only by automated static perimetry (missed on confrontation testing); (2) retinal microaneurysm; and (3) sectoral disc pallor.

Unilateral macular SRF may occur in exudative age-related macular degeneration, optic disc pit [10], focal choroidal excavation [11], choroidal tumours (melanoma, haemangioma, or metastases) [12], dome-shaped macula, tilted disc with inferior staphyloma [13], torpedo maculopathy, and serous maculopathy secondary to retinal pigment epithelium (RPE) dysfunction from confluent drusen etc [14]. Occasionally, pregnancy-related serous maculopathy may be observed [14]. Rarely, aromatic amine-containing hair dyes could cause serous retinopathy[14]. Choroidal folds may be idiopathic, particularly in hyperopic eyes; otherwise, they can be classified into ocular and orbital causes. Ocular causes include hypotony, uveitis, choroidal masses, and PS; while orbital causes comprise TED, IOID, and retrobulbar masses.

Our literature review identified only one study analysing ODR of SRF, vitreous, RPE and RNFL[15]. While this one sole study demonstrated statistically significant ODR differences between VKH versus PS or CSCR, it failed to show significant ODR differences between PS and CSCR. Therefore, distinguishing PS from CSCR still depends primarily on clinical history and physical examination findings.

Funding

Nil.

Sources of support

Nil.

Conflicting interest

Nil.

Presentation at meeting

None.

CRediT authorship contribution statement

Kelly Wai Lam Cheung: Writing – original draft, Methodology, Formal analysis, Data curation. **Sunny Chi Lik Au:** Writing – review & editing, Visualization, Validation, Supervision, Project administration, Methodology, Data curation, Conceptualization.

Declaration of competing interest

Nil.

References

[1] L. Jiangbolati, Y. Gao, Q. Peng, Posterior scleritis misdiagnosed as central serous chorioretinopathy: case report, Photodiagnosis Photodyn Ther 54 (2025 May 29) 104655, https://doi.org/10.1016/j.pdpdt.2025.104655. Epub ahead of print. PMID: 40447063.

[2] P. Zeng, J. Wang, P. Tian, Y.Y. Peng, J.Q. Liang, M. Wang, S.Y. Zhou, Macular and peripapillary optical coherence tomography angiography metrics in thyroid-associated ophthalmopathy with chorioretinal folds, Photodiagnosis Photodyn Ther 42 (2023 Jun) 103146, https://doi.org/10.1016/j.pdpdt.2022.103146. Epub 2022 Oct 6. PMID: 36210040.

[3] S. Lobo, G.J. Lobo, Idiopathic orbital inflammatory disease - a diagnostic dilemma, Rom J Ophthalmol 68 (4) (2024 Oct-Dec) 462–465, https://doi.org/10.22336/rjo.2024.83. PMID: 39936055; PMCID: PMC11809838.

[4] K.K. Lai, F.M. Ali Abdulla Aljufairi, J.U. Sebastian, J.K. Chin, E.K. Choy, A.H. Yiu, A.C. Lee, C.M. Ng, W.W. Yip, A.L. Young, H.K. Yuen, C.C. Tham, C.P. Pang, K. K Chong, Euthyroid graves' ophthalmopathy in a Chinese population: a cross-sectional follow-up study, Eur J Ophthalmol 34 (5) (2024 Sep) 1432–1437, https://doi.org/10.1177/11206721241229470. Epub 2024 Mar 5. PMID: 38444235.

[5] N. Sefi-Yurdakul, True or pseudo optic disc edema: clinically-based approach to the differential diagnosis, Int Ophthalmol 44 (1) (2024 Sep 21) 384, https://doi.org/10.1007/s10792-024-03316-1. PMID: 39305361.

[6] K. Abri Aghdam, A. Aghajani, A. Zand, S. Chaibakhsh, P. Anvari, F.Z. Ijadi, K. Ghasemi Falavarjani, Application of optical coherence tomography angiography in true and pseudo-optic disc swelling, J Ophthalmol 2024 (2024 Sep 30) 1164635, https://doi.org/10.1155/2024/1164635. PMID: 39380943; PMCID: PMC11459931.

[7] Y. Moon, U.S. Kim, J. Han, H. Ahn, H.T. Lim, Clinical and optic disc characteristics of patients showing visual recovery in Leber hereditary optic neuropathy, J Neuroophthalmol 40 (1) (2020 Mar) 15–21, https://doi.org/10.1097/WNO.0000000000000830. PMID: 31609832.

[8] J. Biermann, W.A. Lagrèze, Die randunscharfe Papille: eine diagnostische herausforderung, Klin Monbl Augenheilkd 234 (6) (2017 Jun) 829–850, https://doi.org/10.1055/s-0043-108489. GermanEpub 2017 May 9. PMID: 28486730.

[9] M.B. Rougier, Œdème papillaire bilatéral [Bilateral papilledema], J Fr Ophtalmol 40 (5) (2017 May) 422–429, https://doi.org/10.1016/j.jfo.2017.02.002. French-Epub 2017 May 6. PMID: 28483156.

[10] A. Esmaeil, A. Ali, S. Almutairi, K. Alkandari, R. Behbehani, A. Alali, Congenital optic disc pits and optic disc pit maculopathy: a review, Front Ophthalmol (Lausanne) 3 (2023 Aug 16) 1222979, https://doi.org/10.3389/fopht.2023.1222979. PMID: 38983028; PMCID: PMC11182116.

[11] Y. Gan, G. He, Y. Su, Y. Zhang, X. Zhang, H. Chen, C. Zuo, M. Li, X. Chen, F. Wen, Clinical features of focal choroidal excavation subtypes and their association with choroidal neovascularization, Photodiagnosis Photodyn Ther 53 (2025 Mar 26) 104565, https://doi.org/10.1016/j.pdpdt.2025.104565. Epub ahead of print. PMID: 40154752.

[12] R. Zhang, Y. Li, L. Dong, W. Zhou, W. Wei, Y. Liu, Swept-source optical coherence tomography features of choroidal tumors and tumor mimics: a 5-year retrospective study, Photodiagnosis Photodyn Ther 53 (2025 Mar 27) 104571, https://doi.org/10.1016/j.pdpdt.2025.104571. Epub ahead of print. PMID: 40157474.

[13] H. Liu, F. Meng, B. Wang, J. Li, Y. Xi, Q. Yi, Y. Zhang, Corrigendum to 'Fundus morphological characteristics assesment in high myopia patients with or without posterior scleral staphyloma by optical coherence tomography' [Photodiagnosis and photodynamic therapy, vol. 48. (2024) 1-28/ID: PDPDT_104258], Photodiagnosis Photodyn Ther 49 (2024 Oct) 104297, https://doi.org/10.1016/j.pdpdt.2024.104297. Epub 2024 Aug 20. Erratum for: Photodiagnosis Photodyn Ther. 2024 Aug;48:104258. https://doi.org/10.1016/j.pdpdt.2024.104258. PMID: 39168783.

[14] E.H.C. van Dijk, C.J.F. Boon, Serous business: delineating the broad spectrum of diseases with subretinal fluid in the macula, Prog Retin Eye Res 84 (2021 Sep) 100955, https://doi.org/10.1016/j.preteyeres.2021.100955. Epub 2021 Mar 11. PMID: 33716160.

[15] B. Yargı Özkoçak, S. Ariman, C. Altan, B. Kemer Atik, B. Basarir, Optical density analysis of subretinal fluid in Vogt Koyanagi Harada disease, posterior scleritis and acute Central serous chorioretinopathy, Ocul Immunol Inflamm (2025 Apr 2) 1–8, https://doi.org/10.1080/09273948.2025.2487161. Epub ahead of print. PMID: 40173369.

Kelly Wai Lam Cheung[a], Sunny Chi Lik Au[b,*]

[a] *House Officer, Hospital Authority, Hong Kong*

[b] *Department of Ophthalmology, Tung Wah Eastern Hospital, Hong Kong*

[*] Corresponding author: 9/F, MO Office, Lo Ka Chow Memorial Ophthalmic Centre, Tung Wah Eastern Hospital, 19 Eastern Hospital Road, Causeway Bay, HKSAR, Hong Kong.

E-mail address: kilihcua@gmail.com (S.C.L. Au).